Easy Geometrical Problems — problem 12.3.1.9

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Let a, b, c, d be the sides of a quadrilateral with area A. Prove that

$$\text{(a) } A \leq \frac{ab+cd}{2}, \qquad \text{(b) } A \leq \frac{ac+bd}{2}, \qquad \text{(c) } A \leq \frac{a+c}{2} \cdot \frac{b+d}{2}.$$